

raiz_projeto/docker-compose.yaml

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Define dois “serviços” (máquinas, ou containers)

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Um serviço chamado “database” e outro chamado “application”

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Onde se encontram os Dockerfiles de cada serviço

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O serviço database tem uma variável de ambiente específica

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Se esses serviços devem ser re-startados automaticamente

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Mapeamento de portas <host>:<container>

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Disco: O database possui um volume com o nome “pg_data”, que corresponde ao diretório de Dados do PostgreSQL

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Disco: O application mapeia o diretório "application/src" do projeto para o diretório raiz web do container

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Nome dos containers gerados

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Checagem de Saúde: O serviço é “saudável” (healthy) se e apenas se os comandos presentes em “test” forem executados com sucesso. Em caso de falha tenta, a cada 3s executar novamente por um máximo de 2 tentativas

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O serviço application depende do fato de que o serviço database seja subido com sucesso (“com saúde”, healthy).

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Como o volume de disco do databse ("pg_data") é declarado em uma lista de volumes, ele tem seu estado mantido mesmo após o container ser baixado.

raiz_projeto/database/Dockerfile

```
1 FROM postgres:17
```

Dockerfiles contém instruções de como buildar cada serviço: no caso de database é apenas uma imagem PostgreSQL da versão 17.

raiz_projeto/application/Dockerfile

```
1 FROM php:8.4-apache
2
3 USER root
4
5 RUN apt -y update
6 RUN apt -y install curl sudo vim libnss-wrapper git iputils-ping libicu-dev libonig5 libonig-dev libsodium23 libsodium-dev libxml2-dev libcurl4-openssl-dev libeditreadline-dev libzip5 libzip-dev nodejs npm libdbd-pg-perl postgresql-client libdbd-sqlite3-perl sqlite3 postgresql pgcli libpq-dev
7 RUN docker-php-ext-install bcmath curl exif fileinfo intl mbstring pcntl pdo pdo_pgsql pgsql sodium zip
8
9 RUN a2enmod rewrite
10
11 RUN php -r "copy('https://getcomposer.org/installer', 'composer-setup.php');"
12 RUN php composer-setup.php
13 RUN php -r "unlink('composer-setup.php');"
14 RUN mv composer.phar /usr/local/bin/composer
15
16 RUN composer global require laravel/installer
17 RUN export PATH=${PATH}:/root/.config/composer/vendor/bin
18
19 WORKDIR /var/www/html
20 RUN service apache2 restart
```

No caso da aplicação, builda de uma imagem PHP/Apache (1), instala dependências do Sistema Operacional (5 e 6), extensões php (7), ativa o mod_rewrite do Apache (9), instala o composer e o instalador do laravel (11 a 16) e reinicia o Apache (20).

“Buildando”/Subindo os containers

```
docker compose up -d [--build]
```

Baixando os containers

```
docker compose down
```

Acessando o shell do container

```
docker compose exec -it <nome_serviço> bash
```